

BWS44104: Data Analytics and Management

Synopsis:

This course uses case studies to review the data challenges business confront and how data management and analytics is used to help make sound management decisions.

Course Learning Outcomes (CLO):

At the end of this course, the students will be able to:

CLO1: Recommend an appropriate model for different data types data process and analysis workflow computer systems for each organization; (C4; LOD2; PLO2)

CLO2: Interpret the business significance of the data, what it implies about the business, customers, etc (P4; LOD9; PLO6; SC5)

CLO3: Generating reports on the data, including appropriately constructed graphics and histograms that illustrate important features of the data.(A4; LOD13; PLO9)

TOPIC 1: Introduction to Database Management Systems

Guidelines: Students will learn how to design, organize, and maintain databases using relational models, normalization, and referential integrity.

TOPIC 2: Data Preprocessing

Guidelines: Students will learn how to clean, prepare, and govern data for analysis, including reduction techniques and basic modeling.

TOPIC 3: Understanding Statistics

Guidelines: Students will learn basic statistical concepts such as data sets, size, range, and variation for data analysis.

TOPIC 4: Business Intelligence and Analytics

Guidelines: Students will learn to apply descriptive statistics, visualize data, and explore datasets to generate useful insights.

TOPIC 5: Managerial Decision-Making

Guidelines: Students will learn how to use data queries, models, and reports to support managerial decisions.

Teaching method: Work-based Learning/ Blended Learning

Materials to deliver the course: Lecture slides and notes, Company's data management procedures and governance policies, Case studies and real industry datasets, Data analytics standards or compliance documents.

Expected outcomes: Students will be able to analyze datasets using appropriate techniques, perform data management and analytics tasks confidently with suitable tools, and demonstrate professional values by adhering to governance and ethical standards in practice.

Method of assessment: WBL Logbook /Case Study Analysis / Video Presentation, Practical Observation/ Online Quiz

Assessment: **PLO2:** Cognitive Skill, **PLO6:** Digital Skill, **PLO9:** Personal Skill, **SC5:** Collaboration Competency

 Assessment	Assessor	CLO	PLO	Level	Assessment Tool	%
		CLO1	PLO2	C4	Online Quiz	10

Continuous		CLO3	PLO9	A4	Video Presentation	10
	Industry	CLO1	PLO2	C4	(WBL Progress Logbook Report)	30
		CLO2	PLO6	P4-SC5	Demonstration	20
		CLO3	PLO9	A4	Observation	20
Total						100
Total						100